

Basics, visualizations (BV) - project assignment

Enhanced analysis and visualization for at least three libraries (at least two graph-oriented and at least one hypergraph-oriented), which should include:

- Analysis for at least seven graph datasets (including three real-world, two different random models, and two synthetic) (for graphs)
- Analysis for at least six hypergraph datasets (including three real-world, two different random models, and two synthetic) (for hypergraphs)
- Statistics with histograms (including the one from the Mining Complex Network book, page 18, table 1.1 (for graphs)
- Statistics with histograms (including the one from the Mining Complex Network book (Figure 7.6 and Table 7.7) (for hypergraphs)
- Visualizations (at least 5) (for graphs and hypergraphs)
- Centrality measures defined in the Mining Complex Network book (for graphs)
- Analysis of shortest-path (for graphs)
- Degree correlation (assortativity) analysis (for graphs)

The materials, links, and solutions for the basic assignment may be helpful when building the project results. It would be great to include the libraries that are different from networkx, igraph, or hypernetx.

Examples:

PyVis - graphs: <https://pyvis.readthedocs.io>

- Description: Interactive graph visualization library built on top of **vis.js**. It produces interactive HTML network visualizations with zooming and node interaction.
- interactive graph exploration, Jupyter notebooks, exporting HTML visualizations.

XGI - hypergraphs: <https://xgi.readthedocs.io/en/stable/index.html>.

- Description: A modern Python package designed for analyzing and visualizing higher-order interactions in complex systems (hypergraphs and simplicial complexes).
- Often used in network science research.
- The XGI community also provides an xgi-data repository (with hypergraph repository) <https://github.com/xgi-org/xgi-data>

Graph-tool: <https://graph-tool.skewed.de>

- Description: High-performance graph analysis and visualization library with a C++ core for scalability and efficiency on large graphs.
- Fast, good for large-scale graph analytics.